When ARP Goes Wrong: Threat Hunting ARP Poisoning in Network Traffic

Press enter or click to view image in full size

No.	Time	Source	SrcPort	Destination	DstPort	Protocol
1	2025-07-19 18:11:52.318869340Z	RfLinkTechno_20:1b…		Broadcast		ARP
2	2025-07-19 18:11:52.319706357Z	RfLinkTechno_20:1b…		Broadcast		ARP
3	2025-07-19 18:11:52.320832904Z	RfLinkTechno_20:1b…		Broadcast		ARP
4	2025-07-19 18:11:52.322030188Z	RfLinkTechno_20:1b…		Broadcast		ARP
5	2025-07-19 18:11:52.322931578Z	RfLinkTechno_20:1b…		Broadcast		ARP
7	2025-07-19 18:11:52.325245363Z	RfLinkTechno_20:1b…		Broadcast		ARP
8	2025-07-19 18:11:52.326275505Z	RfLinkTechno_20:1b…		Broadcast		ARP
10	2025-07-19 18:11:52.327331395Z	RfLinkTechno_20:1b…		Broadcast		ARP
11	2025-07-19 18:11:52.328337441Z	RfLinkTechno_20:1b…		Broadcast		ARP
12	2025-07-19 18:11:52.329364053Z	RfLinkTechno_20:1b…		Broadcast		ARP
13	2025-07-19 18:11:52.330323045Z	RfLinkTechno_20:1b…		Broadcast		ARP
14	2025-07-19 18:11:52.331410957Z	RfLinkTechno_20:1b…		Broadcast		ARP
15	2025-07-19 18:11:52.332542456Z	RfLinkTechno_20:1b…		Broadcast		ARP
16	2025-07-19 18:11:52.333514545Z	RfLinkTechno_20:1b…		Broadcast		ARP
17	2025-07-19 18:11:52.395714723Z	RfLinkTechno_20:1b…		Broadcast		ARP
18	2025-07-19 18:11:52.396490395Z	RfLinkTechno_20:1b…		Broadcast		ARP
19	2025-07-19 18:11:52.702848050Z	RfLinkTechno_20:1b…		Broadcast		ARP
21	2025-07-19 18:11:52.703545934Z	RfLinkTechno_20:1b…		Broadcast		ARP
22	2025-07-19 18:11:52.704609671Z	RfLinkTechno_20:1b…		Broadcast		ARP
23	2025-07-19 18:11:52.705687098Z	RfLinkTechno_20:1b…		Broadcast		ARP
24	2025-07-19 18:11:52.706747831Z	RfLinkTechno_20:1b…		Broadcast		ARP
25	2025-07-19 18:11:52.707683674Z	RfLinkTechno_20:1b…		Broadcast		ARP
26	2025-07-19 18:11:52.708728926Z	RfLinkTechno_20:1b…		Broadcast		ARP
27	2025-07-19 18:11:52.709780572Z	RfLinkTechno_20:1b…		Broadcast		ARP
28	2025-07-19 18:11:52.710998280Z	RfLinkTechno_20:1b…		Broadcast		ARP
29	2025-07-19 18:11:52.712140455Z	RfLinkTechno_20:1b…		Broadcast		ARP
30	2025-07-19 18:11:52.712954855Z	RfLinkTechno_20:1b…		Broadcast		ARP
31	2025-07-19 18:11:52.714008998Z	RfLinkTechno_20:1b…		Broadcast		ARP
32	2025-07-19 18:11:52.715066831Z	RfLinkTechno_20:1b…		Broadcast		ARP
33	2025-07-19 18:11:52.716096840Z	RfLinkTechno_20:1b…		Broadcast		ARP

Before we start, it’s important to understand the basics of the ARP protocol.

In a local network, devices do not communicate using IP addresses alone. Actual data transmission at the network level relies on physical (MAC) addresses.

The Address Resolution Protocol (ARP) is responsible for resolving an IP address to its corresponding MAC address so that devices can communicate with each other within the same local network.

for example, when a system wants to communicate with the internet, it must first communicate with the default gateway. Before starting this communication, the system checks its

ARP cache to see whether the MAC address of the gateway is already known.

If the gateway’s MAC address is not present in the ARP cache, the system sends an ARP broadcast request to the entire local network to discover the MAC address of the corresponding device.

In this process, the destination MAC address of the broadcast frame is always set to ff:ff:ff:ff:ff:ff, ensuring that all devices on the local network receive the request.

Please enter or click to view image in full size

arp

No.	Time	Source	SrcPort	Destination	DstPort	Protocol
105	2025-07-19 18:11:57.620326383Z	RfLinkTechno_20:1b…		Broadcast		ARP
106	2025-07-19 18:11:57.771898630Z	RfLinkTechno_20:1b…		Broadcast		ARP
107	2025-07-19 18:11:57.772485114Z	RfLinkTechno_20:1b…		Broadcast		ARP
108	2025-07-19 18:11:57.773826313Z	RfLinkTechno_20:1b…		Broadcast		ARP
109	2025-07-19 18:11:58.002074476Z	RfLinkTechno_20:1b…		Broadcast		ARP
110	2025-07-19 18:11:58.002929892Z	RfLinkTechno_20:1b…		Broadcast		ARP
111	2025-07-19 18:11:58.004233147Z	RfLinkTechno_20:1b…		Broadcast		ARP
112	2025-07-19 18:11:58.232637246Z	RfLinkTechno_20:1b…		Broadcast		ARP
113	2025-07-19 18:11:58.232658563Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP
114	2025-07-19 18:11:58.233240639Z	RfLinkTechno_20:1b…		Broadcast		ARP
115	2025-07-19 18:11:58.234488652Z	RfLinkTechno_20:1b…		Broadcast		ARP
116	2025-07-19 18:11:58.309478803Z	RfLinkTechno_20:1b…		Broadcast		ARP
117	2025-07-19 18:11:58.310060588Z	RfLinkTechno_20:1b…		Broadcast		ARP
118	2025-07-19 18:11:58.311592218Z	RfLinkTechno_20:1b…		Broadcast		ARP
119	2025-07-19 18:11:58.312641475Z	RfLinkTechno_20:1b…		Broadcast		ARP
132	2025-07-19 18:12:14.764781239Z	NetisTechnol_47:fa…		LiteonTechn…		ARP
133	2025-07-19 18:12:14.764803367Z	LiteonTechno_7e:74…		NetisTechno…		ARP
134	2025-07-19 18:12:14.834151274Z	LiteonTechno_7e:74…		NetisTechno…		ARP
135	2025-07-19 18:12:14.836665454Z	NetisTechnol_47:fa…		LiteonTechn…		ARP
142	2025-07-19 18:12:39.705602303Z	SamsungElect_54:6a…		Broadcast		ARP
165	2025-07-19 18:12:54.726244771Z	NetisTechnol_47:fa…		LiteonTechn…		ARP
166	2025-07-19 18:12:54.726266874Z	LiteonTechno_7e:74…		NetisTechno…		ARP
203	2025-07-19 18:13:22.947041669Z	NetisTechnol_47:fa…		LiteonTechn…		ARP
204	2025-07-19 18:13:22.947063678Z	LiteonTechno_7e:74…		NetisTechno…		ARP
29087	2025-07-19 18:13:33.966745761Z	RfLinkTechno_20:1b…		Broadcast		ARP
29088	2025-07-19 18:13:33.967208630Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP
126172	2025-07-19 18:13:40.840028494Z	RfLinkTechno_20:1b…		Broadcast		ARP
126173	2025-07-19 18:13:40.840488144Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP
126174	2025-07-19 18:13:40.993780097Z	RfLinkTechno_20:1b…		Broadcast		ARP

Focusing on ARP requests, we use the arp.opcode == 1 filter to list all ARP request packets.

Press enter or click to view image in full size

arp.opcode==1

No.	Time	Source	SrcPort	Destination	DstPort	Protocol	Host
86	2025-07-19 18:11:55.628523485Z	RfLinkTechno_20:1b…		Broadcast		ARP	
87	2025-07-19 18:11:56.619798146Z	RfLinkTechno_20:1b…		Broadcast		ARP	
88	2025-07-19 18:11:56.620432317Z	RfLinkTechno_20:1b…		Broadcast		ARP	
89	2025-07-19 18:11:56.621453829Z	RfLinkTechno_20:1b…		Broadcast		ARP	
90	2025-07-19 18:11:56.622680574Z	RfLinkTechno_20:1b…		Broadcast		ARP	
91	2025-07-19 18:11:56.624039116Z	RfLinkTechno_20:1b…		Broadcast		ARP	
92	2025-07-19 18:11:56.625054001Z	RfLinkTechno_20:1b…		Broadcast		ARP	
93	2025-07-19 18:11:56.625975056Z	RfLinkTechno_20:1b…		Broadcast		ARP	
94	2025-07-19 18:11:56.627022576Z	RfLinkTechno_20:1b…		Broadcast		ARP	
95	2025-07-19 18:11:56.628219962Z	RfLinkTechno_20:1b…		Broadcast		ARP	
96	2025-07-19 18:11:56.629231068Z	RfLinkTechno_20:1b…		Broadcast		ARP	
97	2025-07-19 18:11:56.630156734Z	RfLinkTechno_20:1b…		Broadcast		ARP	
98	2025-07-19 18:11:56.631209354Z	RfLinkTechno_20:1b…		Broadcast		ARP	
99	2025-07-19 18:11:56.632223161Z	RfLinkTechno_20:1b…		Broadcast		ARP	
100	2025-07-19 18:11:56.850270616Z	RfLinkTechno_20:1b…		Broadcast		ARP	
101	2025-07-19 18:11:56.926975461Z	RfLinkTechno_20:1b…		Broadcast		ARP	
102	2025-07-19 18:11:56.927936929Z	RfLinkTechno_20:1b…		Broadcast		ARP	
103	2025-07-19 18:11:57.618227601Z	RfLinkTechno_20:1b…		Broadcast		ARP	
104	2025-07-19 18:11:57.619089799Z	RfLinkTechno_20:1b…		Broadcast		ARP	
105	2025-07-19 18:11:57.620326383Z	RfLinkTechno_20:1b…		Broadcast		ARP	
106	2025-07-19 18:11:57.771898630Z	RfLinkTechno_20:1b…		Broadcast		ARP	
107	2025-07-19 18:11:57.772485114Z	RfLinkTechno_20:1b…		Broadcast		ARP	
108	2025-07-19 18:11:57.773826313Z	RfLinkTechno_20:1b…		Broadcast		ARP	
109	2025-07-19 18:11:58.002074476Z	RfLinkTechno_20:1b…		Broadcast		ARP	
110	2025-07-19 18:11:58.002929892Z	RfLinkTechno_20:1b…		Broadcast		ARP	
111	2025-07-19 18:11:58.004233147Z	RfLinkTechno_20:1b…		Broadcast		ARP	
112	2025-07-19 18:11:58.232637246Z	RfLinkTechno_20:1b…		Broadcast		ARP	
114	2025-07-19 18:11:58.233240639Z	RfLinkTechno_20:1b…		Broadcast		ARP	
115	2025-07-19 18:11:58.234488652Z	RfLinkTechno_20:1b…		Broadcast		ARP	
116	2025-07-19 18:11:58.309478803Z	RfLinkTechno_20:1b…		Broadcast		ARP	
117	2025-07-19 18:11:58.310060588Z	RfLinkTechno_20:1b…		Broadcast		ARP	

```
▸ Frame 86: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface wlp42s0, id 0
▸ Ethernet II, Src: RfLinkTechno_20:1b:d9 (38:a2:8c:20:1b:d9), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: RfLinkTechno_20:1b:d9 (38:a2:8c:20:1b:d9)
    Sender IP address: 192.168.1.9
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.194
```

An ARP request contains the sender’s IP address and MAC address, along with the target IP address. The target MAC address is unknown at this stage and is set to all zeros, as the purpose of the ARP request is to discover it.

As the next step, we look for the ARP response.

Press enter or click to view image in full size

arp.opcode==2

No.	Time	Source	SrcPort	Destination	DstPort	Protocol	Host
6	2025-07-19 18:11:52.323370335Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
9	2025-07-19 18:11:52.326296217Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
20	2025-07-19 18:11:52.702869999Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
48	2025-07-19 18:11:53.701242050Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
65	2025-07-19 18:11:54.622817393Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
71	2025-07-19 18:11:55.544492275Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
113	2025-07-19 18:11:58.232658563Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
133	2025-07-19 18:12:14.764803367Z	LiteonTechno_7e:74…		NetisTechno…		ARP	
135	2025-07-19 18:12:14.836665454Z	NetisTechnol_47:fa…		LiteonTechn…		ARP	
166	2025-07-19 18:12:54.726266874Z	LiteonTechno_7e:74…		NetisTechno…		ARP	
204	2025-07-19 18:13:22.947063678Z	LiteonTechno_7e:74…		NetisTechno…		ARP	
29088	2025-07-19 18:13:33.967208630Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
126173	2025-07-19 18:13:40.840488144Z	LiteonTechno_7e:74…		RfLinkTechn…		ARP	
126189	2025-07-19 18:13:46.995213343Z	NetisTechnol_47:fa…		LiteonTechn…		ARP	

If nothing appears suspicious at first glance, how can we detect an ARP poisoning attack? What abnormalities should we look for in ARP packets, and how can these abnormalities be identified?

Before hunting attacks, we must know normal ARP behavior:

- ARP requests are broadcast
- ARP replies are unicast
- One IP usually maps to one MAC address
- ARP replies usually follow an ARP request
- ARP traffic volume is low and periodic

Anything that breaks these assumptions is suspicious.

Get BlackMamba’s stories in your inbox

Join Medium for free to get updates from this writer.

Here i am using TSHRK for better understand and better view .

First, we examine ARP requests and replies using tshark, as this provides a clearer understanding of how ARP works.

Press enter or click to view image in full size

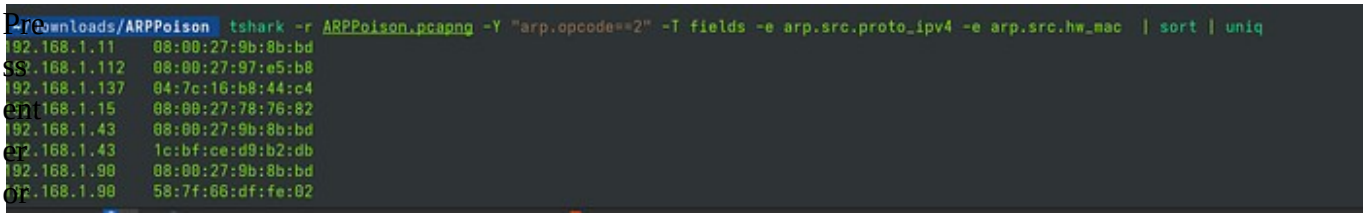
```
/Downloads/ARPPoison  tshark -r ARPPoison.pcapng -Y "arp.src.proto_ipv4 && arp.dst.proto_ipv4"
323 216.143726069 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
324 217.143714891 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
325 218.146208743 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
326 219.394176602 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
327 220.393798034 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
328 221.393834125 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
329 223.643867370 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
330 224.643885798 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
331 225.643879870 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
332 226.703920709 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
333 227.704381068 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
334 228.703940734 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
335 229.764022650 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
336 230.763997807 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
337 231.763980398 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
338 232.824075700 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
339 233.824125947 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
340 234.824107887 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
341 235.884154357 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
342 236.884124374 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
343 237.884154223 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
344 238.944207125 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
345 239.944211126 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
346 240.944237179 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
350 242.004298937 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
353 243.004255189 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
354 244.004308833 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
355 245.064318313 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
361 246.064356348 HuaweiTechno_df:fe:02  Broadcast    ARP    Who has 169.254.40.169? Tell 169.254.100.156
```

Press enter or click to view image in full size

```
2368 801.408024974 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.33? Tell 192.168.1.11
2369 801.418277647 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.27? Tell 192.168.1.11
2370 801.428436594 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.185? Tell 192.168.1.11
2371 801.438574672 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.93? Tell 192.168.1.11
2372 801.448784034 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.47? Tell 192.168.1.11
2373 801.458919647 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.23? Tell 192.168.1.11
2374 801.469101437 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.179? Tell 192.168.1.11
2375 801.479236008 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.214? Tell 192.168.1.11
2376 801.489344110 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.193? Tell 192.168.1.11
2377 801.499448344 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.227? Tell 192.168.1.11
2378 801.509560553 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.253? Tell 192.168.1.11
2379 801.519838353 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.249? Tell 192.168.1.11
2380 801.529994485 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.235? Tell 192.168.1.11
2381 801.540151829 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.152? Tell 192.168.1.11
2382 801.551080573 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.90? Tell 192.168.1.11
2383 801.552027724 HuaweiTechno_df:fe:02  PCSSystemtec_9b:8b:bd  ARP   192.168.1.90 is at 58:7f:66:df:fe:02
2384 801.561569250 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.140? Tell 192.168.1.11
2385 801.571746381 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.16? Tell 192.168.1.11
2386 801.581831269 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.126? Tell 192.168.1.11
2387 801.591974346 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.146? Tell 192.168.1.11
2388 801.602227791 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.26? Tell 192.168.1.11
2389 801.613201139 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.228? Tell 192.168.1.11
2390 801.623352221 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.13? Tell 192.168.1.11
2391 801.634873409 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.75? Tell 192.168.1.11
2395 801.646182628 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.38? Tell 192.168.1.11
2396 801.656297312 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.64? Tell 192.168.1.11
2397 801.666444076 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.6? Tell 192.168.1.11
2398 801.676578457 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.66? Tell 192.168.1.11
2399 801.686771108 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.246? Tell 192.168.1.11
2400 801.697507610 PCSSystemtec_9b:8b:bd  Broadcast    ARP   Who has 192.168.1.176? Tell 192.168.1.11
```

*In this tshark command, **arp.src.proto_ipv4** represents the IP address of the sender generating the ARP packet, while **arp.dst.proto_ipv4** represents the target IP address for which the MAC address is being requested or announced.*

Here, we can clearly observe ARP requests and replies, which helps us understand how the ARP protocol works. Next, we focus on identifying abnormalities by filtering ARP reply packets along with their associated IP and MAC addresses.



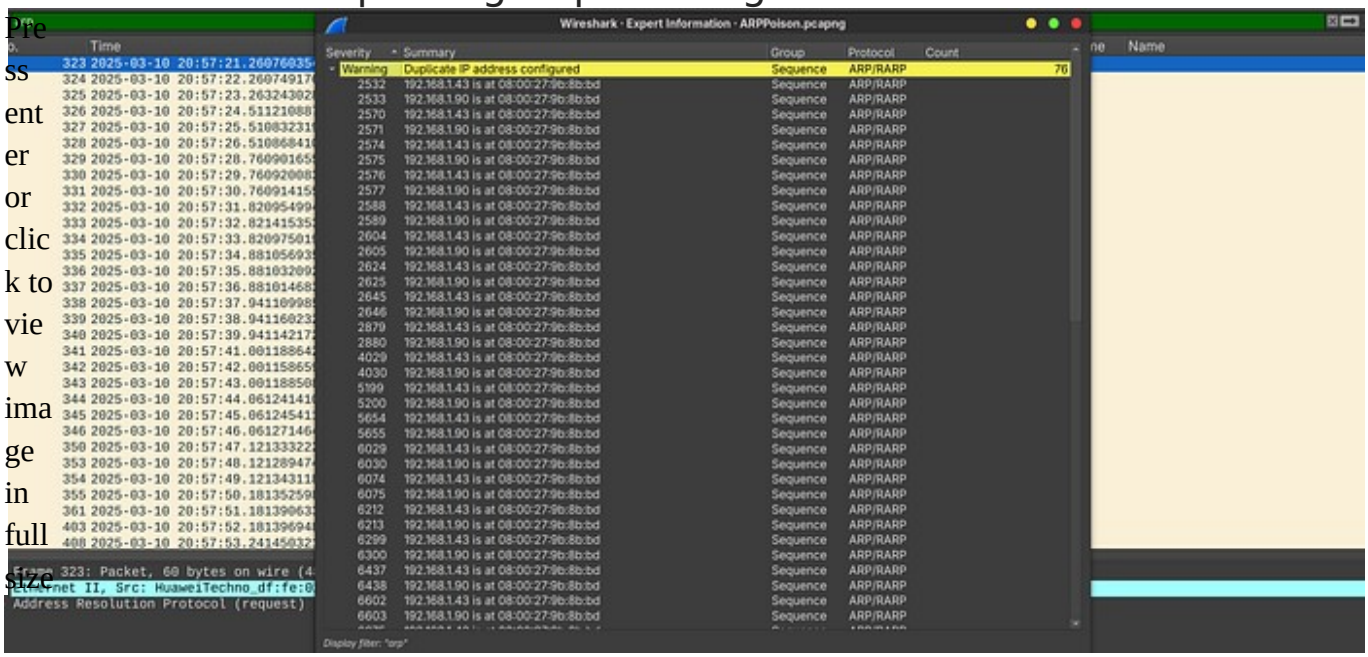
Press enter or click to view image in full size

At this point, we are very close to identifying the issue. As discussed earlier, a MAC address is a unique identifier, and under normal circumstances, each IP address should be associated with a unique MAC address within the same network.

However, in this case, we observe that three different IP addresses are mapped to the same MAC address (08:00:27:9b:8b:bd). This behavior is abnormal and strongly indicates the presence of ARP poisoning on the network.

From the above tshark command, ***arp.src.proto_ipv4*** represents the IP address claimed by the sender of an ARP packet, while ***arp.src.hw_mac*** represents the MAC address making

that claim. Analyzing these fields together helps identify abnormal IP-to-MAC mappings indicative of ARP poisoning.
Another way to confirm ARP poisoning is by using Wireshark’s built-in analysis features.
Open the PCAP file in Wireshark, apply the arp display filter, and then navigate to Analyze → Expert Information.
In the Expert Information window, Wireshark may raise a “Duplicate IP address configured” warning. When this warning is expanded, it reveals that multiple IP addresses are associated with the same MAC address, which is a strong indicator of ARP spoofing or poisoning.



Press enter or click to view image in full size

In this analysis, we started by understanding normal ARP behavior and gradually moved toward identifying abnormalities within ARP packets. By examining unsolicited ARP replies and detecting multiple IP addresses mapped to a single MAC address, we were able to confirm the presence of an ARP poisoning attack.

This investigation demonstrates that ARP poisoning often blends into otherwise normal-looking network traffic and cannot be detected by looking at a single packet in isolation. Instead, recognizing patterns and inconsistencies over time is critical.

Since ARP lacks built-in authentication, it remains a viable attack vector in many environments, particularly on public or unsecured networks. For defenders, continuous monitoring of ARP traffic and enforcing network-level protections are essential steps toward mitigating these attacks.